Title: Diagnostic and prognostic potential of microRNA clusters in human lung cancers.

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Abstract:

Introduction: Lung cancer is the leading cause of cancer-related deaths worldwide, and earlier diagnosis can reduce mortality. MiRNA clusters, which co-express and target mRNAs, may be valuable biomarkers. This study explores their diagnostic and prognostic potential in lung cancers. Materials and Methods: We analyzed miRNA and mRNA sequencing data from 663 primary tumors and 89 normal controls using DeSEQ2 for differential expression. The dataset was split into 70% for training and 30% for testing, employing k-fold cross-validation (k=5) to minimize overfitting. Various machine learning algorithms were tested for identifying prognostic miRNA clusters, with hyperparameters optimized through grid search. Results: Differential expression analysis revealed 266 (20.04%) and 492 upregulated (37.10%), 177 (13.38%), and 135 (10.18%) downregulated miRNAs in lung adenocarcinoma and squamous cell carcinoma. The SVM model emerged as the top performer for lung squamous cell carcinoma, achieving 99% accuracy with the cluster chrX_10. For lung adenocarcinoma, cluster chr17_6 performed well but lacked specificity. Decision trees and logistic regression were less effective, though cluster chr17_14 showed significance for lung squamous cell carcinoma. Conclusions: Analysis of miRNA clusters indicates their potential as diagnostic and prognostic biomarkers in lung cancer, particularly clusters like chrX_10 for lung squamous cell carcinoma and chr17_6 for lung adenocarcinoma. Challenges persist regarding overfitting and false positives, necessitating further validation.

Topic & Research Question:

Lung cancers pose a significant threat to global economic and social development, with current treatment strategies insufficient to meet the rising demands for diagnosis and care. The heterogeneous nature of the disease and the complexities of histological grading, tumor staging, and molecular phenotypes hinder the discovery of effective therapeutic targets. While many studies examine individual microRNAs (miRNAs) in cancer, there is a lack of research on miRNA clusters. These clusters can act as tumor suppressors or oncogenes, potentially targeting similar genes or pathways to influence carcinogenesis in a coordinated manner.

This study aims to answer the question: *Do miRNA clusters have significant potential as diagnostic and prognostic biomarkers across different lung cancer types?* A thorough

analysis of their clinical relevance is essential before advancing therapeutic approaches.

Background:

Lung cancer is the leading cause of death globally, estimated to account for 2.4 million new cases and 1.8 million deaths in 2022, presenting a significant public health challenge (1). Primarily, there are two main types of lung cancer - non-small cell lung cancer (NSCLC), which accounts for about 85% of all lung cancer cases, and small cell lung cancer (SCLC), which accounts for the remaining 15% of cases (2). NSCLC is further categorized into adenocarcinoma (LUAD), squamous cell carcinoma (LUSC), adenosquamous cell carcinoma, large cell carcinoma, and undifferentiated NSCLC. LUAD represents the most common of all NSCLC cases (40%), arises in mucus-making gland cells in the airway lining, and is stain-positive for thyroid transcription factor I (TTF-I) or Napsin A (3). LUSC, representing 30% of all NSCLC cases, arises in flat cells covering the surface of the airway and stain positive for p63/p40, cytokeratin CK5/6, and TTF-1 negative. Beyond the loss of lives, lung cancer is also a significant threat to global economic and social development, straining healthcare systems and affecting millions of families. Early detection and treatment can improve management; however, challenges such as late-stage diagnoses, inaccessible testing, and high treatment costs hinder this process, leading to poor outcomes. These challenges underscore the need for affordable and accessible strategies with high accuracy, specificity, and sensitivity for better disease management.

Cancer is a disease marked by an accumulation of genetic and epigenetic changes that cause cells to function abnormally. Genome instability is a “facilitating characteristic” of most cancer cells that helps to fuel the hallmarks of cancer. It refers to the increased frequency of mutations in the genome, such as single nucleotide variations (SNVs), small insertions and deletions (indels), copy number variations (CNVs), and structural rearrangements, and leads to gross genetic alterations in cancer (4,5). Genetic and epigenetic changes, such as DNA mutations, methylation, and histone modifications, play vital roles in cancer. For example, TP53 mutations disrupt tumor suppression (6), while DNA hypermethylation silences key tumor-suppressor genes (7–9). These mechanisms influence gene expression and cellular behavior, providing targets for therapies.

Among these modulators, microRNAs (miRNAs) have emerged as critical players. Discovered in Caenorhabditis elegans (C.elegans) by Ambros and colleagues in 1993, miRNAs range typically 20-22 nucleotide (nt) in length and regulate gene expression post transcription by binding to the 3’-UTR of target mRNAs (10,11). When deregulated in cancers, they impact key pathways such as cell proliferation, cell cycle,

apoptosis, migration, and invasion (12–16). Due to their tissue specificity, stability, and presence in body fluids, miRNAs are considered ideal candidates as cancer biomarkers.

MiRNAs are significantly clustered at certain genomic regions (17–19), suggesting they might be transcribed as a single polycistron (20,21). Over 20% of known microRNAs (miRNAs) are organized into genomic clusters (22). These clusters consist of genes located within 10 kb of each other on the same chromosome and strand (17). Research indicates clustered miRNAs often co-expressed and target similar pathways (23–26). The miR-17~92 cluster at 13q31, containing six miRNAs, is amplified in various cancers like B-cell lymphomas and lung cancer, showing a threefold increase in triple-negative breast cancer (27–30). Its expression is regulated by the proto-oncogene c-Myc, affecting cell proliferation and apoptosis. Although there has been extensive research on miRNAs implicated in cancers, studies of large miRNA clusters in cancer are rare.Indeed, a unified analysis that explores the clinical significance of miRNA clusters across various cancer types is important for its in-depth knowledge before venturing into therapeutics.

Hypotheses:

We hypothesize that specific miRNA clusters play crucial roles in cancer progression and have significant potential as biomarkers for cancer diagnosis and prognosis. Due to their clustered nature and co-expression, these miRNAs likely target common cancer-related pathways, enabling them to influence tumor development and progression in a coordinated fashion. By studying these clusters across various cancer types, we aim to demonstrate that miRNA clusters offer high diagnostic accuracy and prognostic value, facilitating earlier detection and improved patient outcomes.

Methods:

Genomic Annotation of Clustered miRNAs: MiRBase is a repository for miRNA sequences and annotations that provides information on miRNA clusters and their genomic locations, aiding in their identification and classification for specific analysis. The genomic locations of the miRNAs were downloaded from miRBase v22 (31). MiRNAs within 10 kb of distance were clustered on the same chromosome and strand. We identified 481 miRNAs to be nested within 159 miRNA clusters.

Downloading and Preparing TCGA Data: The Cancer Genome Atlas (TCGA), a collaboration between the National Cancer Institute (NCI) and the National Human Genome Research Institute (NHGRI), offers high-throughput sequencing data on miRNA expression from 33 primary human cancers and matched controls. The miRNA expression data from the Illumina platform, aligned to hg38, was downloaded using the R package TCGAbiolinks v2.30.4 (32) for human lung cancers.

Preprocessing and Normalization of the miRNA Expression Matrix: The miRNA expression matrix was filtered and normalized to remove technical variability. Samples with less than 60% CPE were removed per the recommended threshold (33,34). To reduce the memory size and to increase the speed and power of the test functions, miRNAs with a total read count < 10 were removed.

Normalization to Remove Technical and Confounding Variability: The miRNA expression values were normalized for differences in sequencing depth and library composition using the "variance Stabilizing Transformation" function of the R package, DeSEQ2 v1.42.1 (35).

Differential Expression of miRNA Clusters: Differential expression was studied using DESeq2 v1.42.1 (35) based on the negative binomial distribution. The statistically significant differences in expression between tumor and normal samples were determined using the following threshold of log2 fold change (log2FC) and adjusted p-value (adj.p), upregulation (log2FC ≥0.6, adj.p ≤0.05) and downregulation (log2FC ≤-0.6, adj.p ≤0.05). The significance of the differential expression of clustered miRNAs was determined using a test for independence in a 2×2 (contingency) table. The differential expression of the miRNA cluster was compared to a standard defined by the complement of that cluster, i.e., miRNAs not in the cluster. The p-value for the over-representation of the clustered miRNAs among the differentially expressed genes was determined using a proportion test. Statistical significance was assumed when the p-value was ≤ 0.05.

Classification Based on miRNA Clusters Using Machine Learning Algorithms: The dataset was split into training (70%), and testing (30%) sets for an unbiased model evaluation using the R package, Caret v6.0, while subsequent analyses were conducted in Python. K-fold cross-validation (k=5) was applied to the training set to enhance reliability and reduce overfitting. Various machine learning algorithms, including Support Vector Machines (SVM), Random Forest, and Logistic Regression, were tested for classifying cancer types and identifying prognostic miRNA clusters, with hyperparameters optimized through grid search. Model performance was assessed based on accuracy. The final model performance was evaluated using AUC-ROC, sensitivity, and specificity metrics on the test set.

Data Processing and Statistical Analysis: Statistical analyses mentioned above were performed using programming software, R Foundation For Statistical Computing, Vienna, Austria (version 4.3.3), and Python. Statistical significance was assumed if $p <$

0.05. Scripts used for data download, analysis, and visualization have been included in the Appendix.

Results:

Differential expression analysis identified significant miRNAs in lung cancers.

Pre-filtering, the dataset included 519 Primary Tumor samples and 46 Solid Tissue Normal samples for TCGA-LUAD, 478 Primary Tumor samples, and 45 Solid Tissue Normal samples for TCGA-LUSC. After filtering, the sample counts were reduced to 314 Primary Tumor samples, 46 Normal samples for TCGA-LUAD, 349 Primary Tumor samples, and 45 Normal samples for TCGA-LUSC. After filtering, the total number of miRNAs decreased from 1881 to 1327 for TCGA-LUAD and from 1881 to 1326 for TCGA-LUSC. Next, we performed principal component analysis (PCA) (Figure 1A). For TCGA-LUAD, Principal Component 1 (PC1) explained 13% of the variance, while Principal Component 2 (PC2) explained 8%. For TCGA-LUSC, PC1 accounted for 14% of the variance, and PC2 accounted for 7%. The separation along PC1 was more pronounced for LUSC, although one Normal sample clustered with the Tumor group. Differential expression analysis revealed 266 (20.04%) and 492 upregulated (37.10%), 177 (13.38%), and 135 (10.18%) downregulated miRNAs in LUAD and LUSC (Figure 1B).

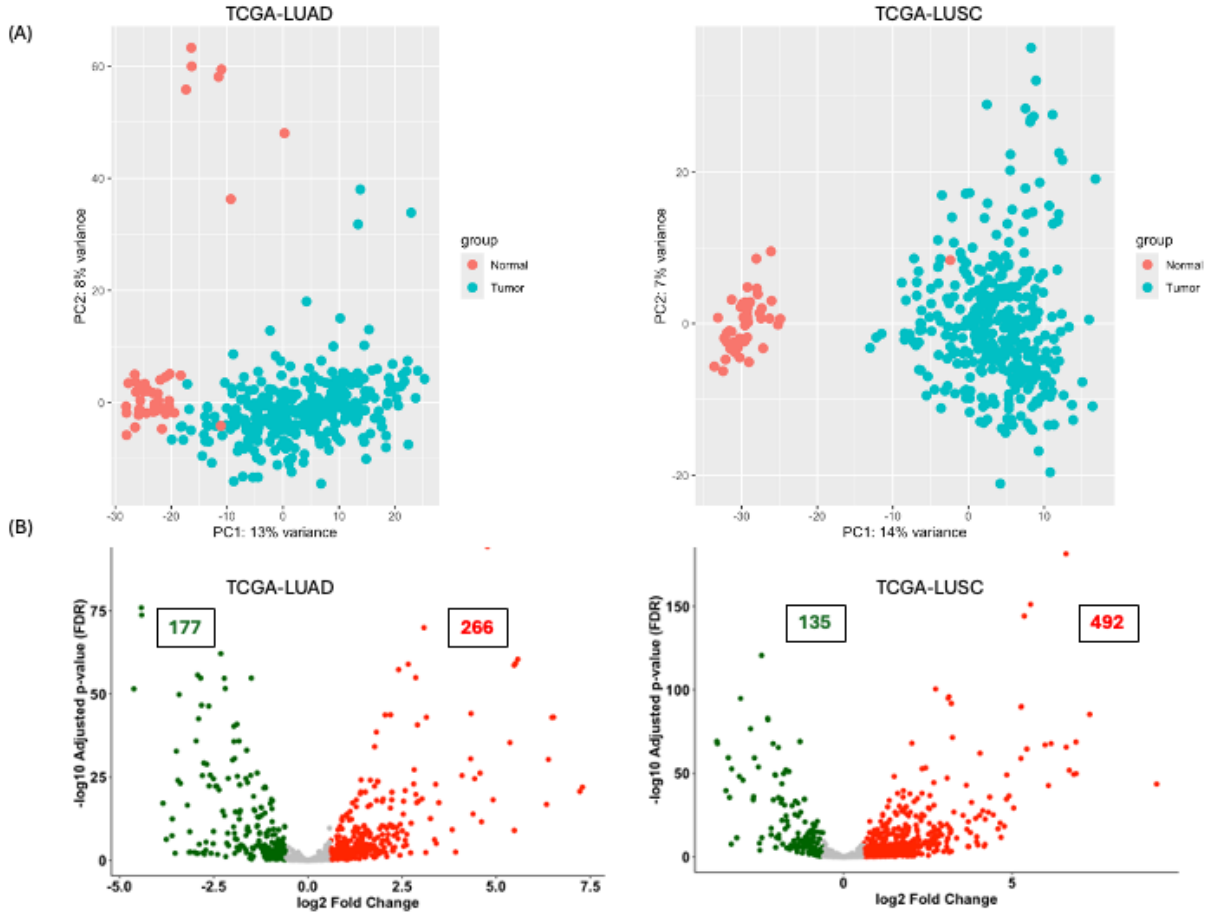

Figure 1. (A) PCA plot of 500 top features by variance where each dot represents a sample condition. (B) *Volcano plot showing differential expression.*

Performance of machine learning models in classifying lung cancer miRNA clusters.

Following preprocessing and normalization, annotations from MiRBase were downloaded to consolidate miRNA genes into their respective clusters. Most of the clusters had less than four miRNA members. For enough statistical power, we looked at clusters with at least four members. In total, each lung cancer dataset contained 32 miRNA clusters. Train and test were split into training (70%) and testing (30%) for each miRNA cluster model.

Support Vector Machine: Support vector machine was implemented with grid search hyper tuning for C and sigma parameters. A k-fold validation = 5 was implemented on the training set. Table 2 is the top 10 cluster classifiers based on accuracy for LUAD:

	Model	Accuracy	F1_Score	Best_C	Best_Sigma
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
Precision31	Cluster chrX_9	0.9626168	0.8333333	8.00000	2.000000e+00
Precision7	Cluster chr17_6	0.9532710	0.7826087	16.00000	1.250000e-01
Precision1	Cluster chr13_2	0.9252336	0.6000000	32.00000	2.500000e-01
Precision3	Cluster chr14_3	0.9252336	0.5555556	32.00000	5.000000e-01
Precision22	Cluster chr9_5	0.9252336	0.6666667	16.00000	2.500000e-01
Precision29	Cluster chrX_2	0.9065421	0.5000000	32.00000	5.000000e-01
Precision6	Cluster chr17_14	0.8971963	0.3529412	32.00000	4.000000e+00
Precision11	Cluster chr19_8	0.8971963	0.4210526	16.00000	2.000000e+00
Precision21	Cluster chr9_4	0.8878505	0.3333333	8.00000	1.562500e-02
Precision30	Cluster chrX_6	0.8878505	0.2500000	0.50000	6.250000e-02

Table 1. LUAD top performing cluster SVM models accuracy results.

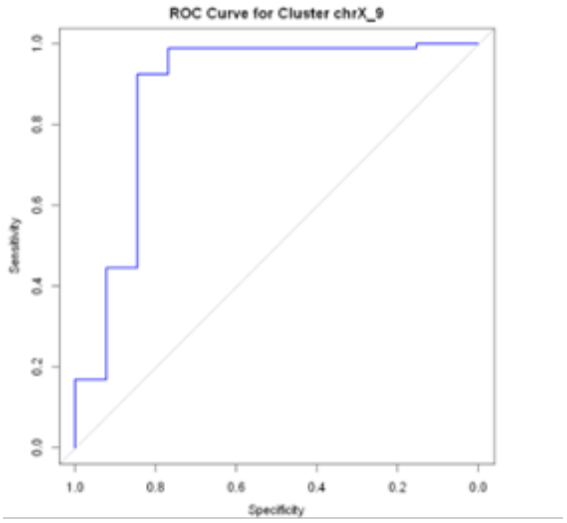

Figure 2. ROC curve cluster "chrX_9"

Despite hyper tuning to improve instrument performance, the top highly accurate cluster models have low F1 scores. This model struggles to classify tumor adenocarcinomas. MiRNAs in cluster chrX_9 is the best model in classifying positive incidences. This model shows high sensitivity with a significant false positive rate.

SVM performed on LUSC provided cluster models with higher accuracy and F1 scores in Table 3. Clusters of high accuracy shared between the groups would not be ideal candidates for the classification of tumors.

	Model	Accuracy	F1_Score	Best_C	Best_Sigma
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
Precision23	Cluster chrX_10	0.9829060	0.9166667	32.00000	3.125000e-02
Precision29	Cluster chrX_2	0.9829060	0.9166667	8.00000	7.812500e-03
Precision5	Cluster chr17_14	0.9743590	0.8800000	16.00000	6.250000e-02
Precision18	Cluster chr7_8	0.9743590	0.8888889	32.00000	1.250000e-01
Precision30	Cluster chrX_6	0.9658120	0.8461538	8.00000	3.125000e-02
Precision6	Cluster chr17_6	0.9572650	0.8000000	0.03125	3.125000e-02
Precision3	Cluster chr14_3	0.9487179	0.7500000	32.00000	6.250000e-02
Precision17	Cluster chr7_6	0.9401709	0.6666667	32.00000	7.812500e-03
Precision	Cluster chr11_3	0.9230769	0.5714286	32.00000	1.250000e-01
Precision14	Cluster chr22_6	0.9230769	0.5263158	8.00000	6.250000e-02

Table 2. LUSC top performing cluster SVM models accuracy results.

Cluster chrX_10 is the most promising in classifying this cancer type, with accuracies more significant than 98% and F1 scores greater than 0.9. The LUSC data set tends to show higher overfitting based on the C and sigma parameters. This limits how variable samples are allowed to perform a reasonable classification. Further experimentation with more variable samples, such as having various stages of tumor progression, is needed to improve the models as prognostic biomarkers. However, at 98% accuracy, cluster “chrX10” and high sensitivity, this cluster is an optimal candidate for a diagnostic biomarker.

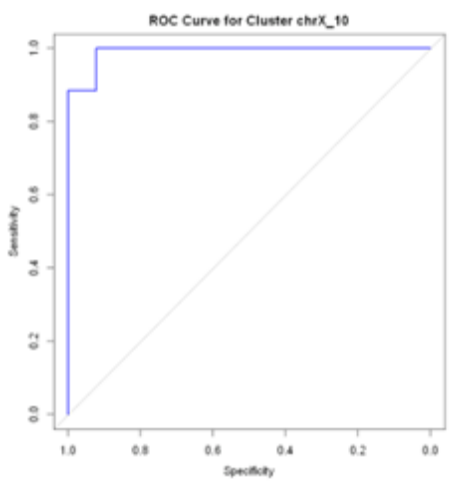

Figure 3. ROC curve cluster "chrX_10"

Decision Tree: Decision tree was used with a k-fold validation = 5 with grid search optimization on the complexity parameter. Trained on LUAD data, the decision tree showed a significant decrease in the cluster model's ability to classify positive incidences compared to the SVM model. MiRNA cluster "chr17_6" are highly scored classifier models in the decision tree and SVM. This suggests that this cluster expression is distinctly different between normal and tumor.

	Model	Accuracy	F1_Score	AUC	Best_CP
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
Precision7	Cluster chr17_6	0.9345794	0.6666667	0.7639116	0.10
Precision18	Cluster chr7_8	0.9345794	0.7407407	0.8903437	0.03
Precision3	Cluster chr14_3	0.9252336	0.5555556	0.7937807	0.10
Precision23	Cluster chrX_10	0.9158879	0.5714286	0.8506547	0.10
Precision12	Cluster chr19_9	0.9065421	0.3750000	0.7176759	0.03
Precision6	Cluster chr17_14	0.8971963	0.5217391	0.7917349	0.03
Precision21	Cluster chr9_4	0.8971963	0.5217391	0.8248773	0.01
Precision30	Cluster chrX_6	0.8971963	0.3529412	0.6423895	0.07
Precision29	Cluster chrX_2	0.8878505	0.5384615	0.8117840	0.10
Precision31	Cluster chrX_9	0.8878505	0.4545455	0.6853519	0.07

Table 3. LUAD top performing cluster decision tree models accuracy results.

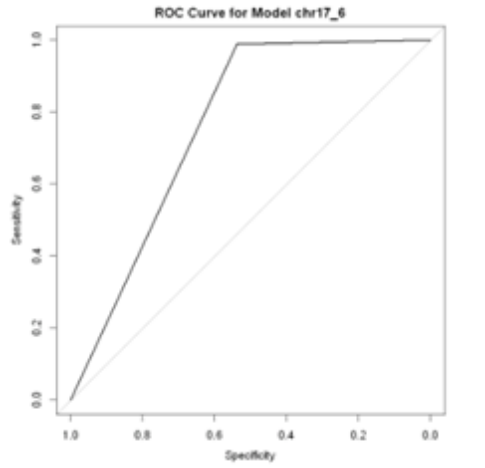

Figure 4. ROC curve cluster "chr17_6".

The decision tree performed poorly in classifying positive incidences. Despite having high complexity (overfitting), the decision tree is not ideal for the classification of LUAD.

LUSC performed decently, although high rates of false positives in the algorithms were detected. SVM performed better on specific clusters. However, both models performed similarly for cluster “chr17_14”. These further increases cluster “chr17_14” diagnostic power with multiple models confirming its significance. It is important to note that the high complexity indicates significant overfitting to the data.

	Model	Accuracy	F1_Score	AUC	Best_CP
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
Precision5	Cluster chr17_14	0.9743590	0.88000000	0.9182692	0.10
Precision6	Cluster chr17_6	0.9572650	0.80000000	0.8750000	0.10
Precision30	Cluster chrX_6	0.9572650	0.80000000	0.8953402	0.10
Precision23	Cluster chrX_10	0.9401709	0.77419355	0.9726331	0.05
Precision	Cluster chr11_3	0.9316239	0.60000000	0.7200444	0.01
Precision17	Cluster chr7_6	0.9316239	0.55555556	0.7174556	0.03
Precision3	Cluster chr14_3	0.9230769	0.47058824	0.6068787	0.09
Precision22	Cluster chr9_5	0.9230769	0.60869565	0.8472633	0.01
Precision11	Cluster chr19_9	0.9145299	0.61538462	0.8731509	0.05
Precision14	Cluster chr22_6	0.9059829	0.56000000	0.7451923	0.07

Table 4. LUSC top performing cluster decision tree models accuracy results.

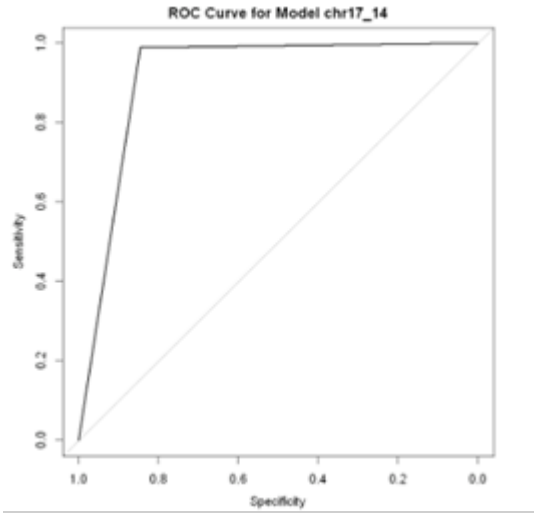

Figure 5. ROC curve cluster "chr17_14".

Logistic Classifier: Logistic classifier was used with a k-fold validation = 5. Optimization was attempted using boosting but performed worse pre-optimization. The pre-optimized model performed similarly to the decision tree, making SVM more suitable for LUAD classification. Cluster "chr17_6" is a top hit in accurately classifying LUAD. Despite cluster "chr17_6" showing high accuracy in many models, this cluster was also found to have a high accuracy in the LUSC data, suggesting that this cluster is distinctive in tissue type. Further experimentation is required to determine if there are significant differences in expression between the cancers. This omits cluster "chr17_6" from being a unique potential biomarker for LUAD.

	Model	Accuracy	F1_Score	AUC
	<chr>	<dbl>	<dbl>	<dbl>
Precision7	Cluster chr17_6	0.9439252	0.7000000	0.9615385
Precision17	Cluster chr7_6	0.9345794	0.7200000	0.9509002
Precision31	Cluster chrX_9	0.9252336	0.6666667	0.9672668
Precision18	Cluster chr7_8	0.9158879	0.7272727	0.9648118
Precision13	Cluster chr1_1	0.9065421	0.7058824	0.9615385
Precision1	Cluster chr13_2	0.8971963	0.3529412	0.8690671
Precision3	Cluster chr14_3	0.8971963	0.5600000	0.7426350
Precision22	Cluster chr9_5	0.8878505	0.2500000	0.8207856
Precision23	Cluster chrX_10	0.8878505	0.4545455	0.9353519
Precision30	Cluster chrX_6	0.8878505	0.4000000	0.8788871

Table 5. LUAD top performing cluster logit models accuracy results.

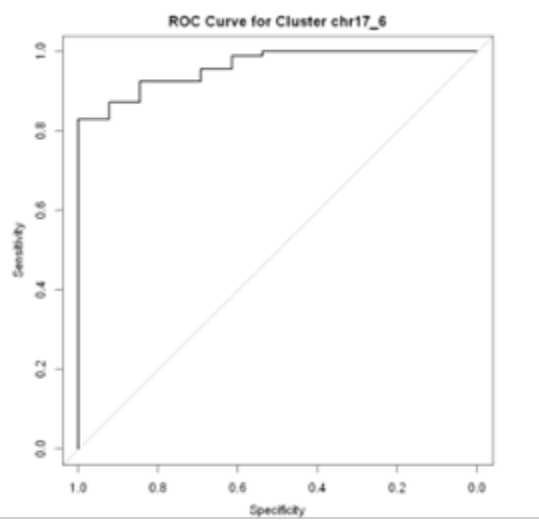

Figure 6. ROC curve cluster "chr17_6".

Logit regression performed the best within the LUSC dataset, and the top cluster models of interest are cluster "chrX_10".

	Model	Accuracy	F1_Score	AUC
	<chr>	<dbl>	<dbl>	<dbl>
Precision23	Cluster chrX_10	0.9914530	0.9600000	1.0000000
Precision5	Cluster chr17_14	0.9743590	0.8888889	0.9585799
Precision6	Cluster chr17_6	0.9743590	0.8888889	0.9630178
Precision29	Cluster chrX_2	0.9743590	0.8695652	0.8846154
Precision18	Cluster chr7_8	0.9658120	0.8571429	0.9807692
Precision30	Cluster chrX_6	0.9658120	0.8461538	0.9918639
Precision14	Cluster chr22_6	0.9487179	0.7000000	0.9622781
Precision	Cluster chr11_3	0.9401709	0.7200000	0.9748521
Precision17	Cluster chr7_6	0.9230769	0.5714286	0.9548817
Precision21	Cluster chr9_4	0.9230769	0.5714286	0.9408284

Table 6. LUSC top performing cluster logit models accuracy results.

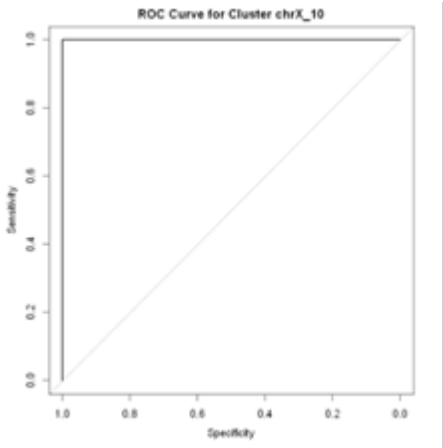

Figure 7. ROC curve cluster "chrX_10".

Cluster model "chrX_10" performs at 99% accuracy, making it a strong potential for diagnostics. SVM confirms this cluster as among the top performers in classification.

Discussion and Conclusion:

The analysis of lung cancer datasets revealed distinct differences in the classification performance of various machine learning models for LUSC and LUAD. Overall, the LUSC dataset showed promising classification results across all algorithms, consistently demonstrating high accuracy. Notably, cluster "chrX_10" referred to as miR-424(322)/503 emerged as a top-performing classifier, achieving an impressive accuracy of 99% with logistic regression models. This cluster also performed well with support vector machine (SVM) models, indicating its strong potential as a reliable biomarker for LUSC. This cluster regulates fundamental tumorigenic processes such as cell proliferation, cell cycle distribution, and epithelial-to-mesenchymal transition (EMT), though with paradoxical roles and complex mechanisms (36). In addition, the cluster acts as a tumor suppressor in breast cancer and links mammary epithelial involution, tumorigenesis, and chemoresistance (37).

In contrast, the LUAD dataset generally exhibited lower classification performance than LUSC. The most accurate model for LUAD was cluster "chrX_9," corresponding to miR-106a-363, which utilized SVM. Although it did not reach the high accuracy in LUSC, it performed well in sensitivity. The optimization of C and sigma parameters allowed the SVM model for LUAD to generalize better, making it more adaptable to diverse inputs and potentially valuable for broader applications. The miR-106a-363 cluster containing hsa-miR-20b-5p and hsa-miR-363-3p affect the expression of key tumor suppressor genes, including PTEN and BIM, and have been shown to modulate the survival of T-cell acute lymphoblastic leukemia (T-ALL) cells in vitro (38). In addition, miR-106a is implicated in lung cancer, especially in cases with bone

metastasis, where it regulates autophagy-related cell death and epithelial-mesenchymal transition (EMT) by targeting TP53INP1 (39).

One notable finding was the shared top-performing cluster between both cancer types: cluster “chr17_6,” which corresponds to the miR-1792 cluster. This cluster was consistently identified as significant across multiple models, making it an essential candidate for further investigation. Its recurrent prominence in LUAD and LUSC suggests it may be critical in lung diseases. This cluster is amplified in various cancers, including B-cell lymphomas and lung cancer, and is frequently translocated in T-cell acute lymphoblastic leukemia, contributing to the overexpression of the cluster in different malignancies (27–29). Moreover, deep sequencing data from triple-negative breast cancer samples have shown a threefold increase in the expression of the miR-1792 cluster (40). The proto-oncogene c-Myc, often upregulated in cancers, represses the transcription of the miR-1792 cluster, thereby regulating cell proliferation and apoptosis (41). These findings suggest that miR-17~92 (cluster “chr17_6”) could be a critical driver of tumorigenesis in lung cancers, warranting further investigation into its biological function and diagnostic potential.

Interestingly, miR-106a-363 is a paralogue of miR-17-92, likely arising from gene duplication events in early vertebrates (42). These paralogues share highly similar, and sometimes identical, miRNAs, suggesting they regulate overlapping sets of target genes. Their shared functions underscore the complexity of miRNA regulation and highlight their potential as diagnostic or therapeutic targets in lung cancer.

Limitations and Other Interpretations:

Data Quality Variability: Results from this analysis are foundations for hypothesis generation to confirm these distinct cluster profiles between both diseases. Since datasets are analyzed using different protocols and platforms, conducting studies to minimize this technical variability is essential.

Tumor Heterogeneity: Many models showed a lack of generalization and high overfitting. To overcome this, future studies with larger datasets are required to train the models properly, increasing the classification integrity of the model. To optimize the model for prognosis, controlled experiments with samples at different stages of progression are necessary to account for the variety most likely attributed to the dataset.

Causality vs. Association: MiRNA clusters may correlate with cancer, but causal links need experimental validation. Experimental validation through functional studies, such as knockdown or overexpression models, is required to determine whether these

clusters actively drive tumor development or are merely byproducts of cancer-related processes.

Interaction with Other Genetic Factors: Due to the nature of miRNA cluster interaction with other genomic factors, experimentation is required to validate the results from this analysis. Elucidating mechanistic and pathway relations of significantly expressed miRNA clusters can be the basis for understanding tumor progression.

Future Research Directions:

To address overfitting in these models, future studies will supplement data from other databases to gain a larger sample size and diversity. With top cluster model hits, future experiments were conducted using batch effect correction to investigate further the role these clusters have in each subset of lung cancer. Multi-cohort analysis can potentially define more significant clusters to reach broader across multiple populations. Larger data sets can resolve the overfitting issue and elucidate clusters that may co-express in specific lung cancer types. The potential to have this model used for prognosis is contingent on working on datasets with various subtypes and stages. Added sub-group analysis adds power to the model in determining patient outcomes.

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